Ali Holmov*

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EDUCATION

M.Sc. Computer Science | Technical University Munich | GPA: 1.6 (1.0 = highest) 2025–now
B.Sc. Computer Science | University of Marburg | GPA: 1.3 (1.0 = highest) 2021–2025
B.A. Philosophy | Saint Petersburg University | GPA: 4.7 (5.0 = highest) 2016–2020

RESEARCH EXPERIENCE

Student Researcher | Explainable ML Munich (Prof. Akata) | Munich, Germany Mar. 2026 – Sep. 2026
- Topic: Isolating latent belief subspace of user representations via self-distillation from unlabeled chat data.

B.Sc. Thesis Student | Oxford (Dr. Schoots, with Prof. Seifert) | Remote Apr. 2025 – Sep. 2025
- Grade: 1.0, Topic: Investigating Pruning-Based Approaches for Reversing ROME Edits.
- Trained a single binary mask across 3,000 independently edited MLP weight matrices to isolate shared parameter subspaces in edited LLMs.

Student Researcher | XAI Group, hessian.AI (Prof. Seifert) | Marburg, Germany Nov. 2023 – Sep. 2025
- Topic: Knowledge editing in LLMs (ACL 2026 paper).

Research Intern in AI Safety | King's College London | London, UK Aug. 2023 – Oct. 2023
- Topic: Lottery Ticket Hypothesis & Effective Density of Feed-Forward models (ECAI 2024 paper).
- Analyzed network overparameterization regimes under magnitude pruning, demonstrating scale-invariant effective density in FNNs.

PUBLICATIONS

CONFERENCE PAPERS

User Model Extraction via Belief Self-Distillation 2026
Ali Holmov, Yiran Huang, Kirill Bykov, Zeynep Akata
Preprint, under review | arXiv | code

One Mask to Rule Them All: On Hidden Facts after Editing and How to Find Them 2026
Ali Holmov, Paul Youssef, Nandi Schoots, Christin Seifert
Findings of ACL | paper | code

The Propensity for Density in Feed-forward Models 2024
Nandi Schoots, Alex Jackson, **Ali Kholmovaia**, Peter McBurney, Murray Shanahan
ECAI | paper

JOURNAL ARTICLES

Challenges for Human Oversight in Detecting System Malfunctions in AI-Assisted Medical Diagnostics 2026
Nadine Schlicker, Constanze Tepohl, **Ali Kholmovaia**, Markus Langer, Noëlle Steffens, Ivica Grgic, Niels van Berkel, Andreas Mayr, Martin Hirsch
npj Digital Medicine

Physicians Trust AI Less in Complex Cases, but Rely on It Just as Much 2026
Nadine Schlicker, Noëlle Steffens, **Ali Kholmovaia**, Orestis Loukas, Leander E. Melms, Martin C. Hirsch, Markus Langer
Technology, Mind, and Behavior

*Using Holmov, a simplified version of my surname (Kholmovaia).

SCHOLARSHIPS & AWARDS

Friedrich-Ebert-Stiftung Scholarship merit-based full scholarship for academically gifted students with high commitment to democratic values — 2025–2027

Hessian State Teaching Award for Student Initiative AI-Lab: AI & Society, Hessian Ministry of Science — 2025

DAAD Award for Outstanding International Students at German Universities University of Marburg — 2023

DAAD-STIBET Scholarship for Outstanding Engagement during the War in Ukraine University of Marburg — 2022

INDUSTRY EXPERIENCE

ML Engineer | Helmit, part-time | Munich, Germany — Nov. 2025 – now

- Owning chat-safety classification in production: 1.1M chat analyses at 63% higher recall than any commercial moderation API benchmarked.
- Developed an on-device content moderation model using reasoning distillation, achieving 93% parameter reduction while outperforming the production model across all benchmarks.

AI Engineer | AUDI, internship | Ingolstadt, Germany — Apr. 2025 – Sep. 2025

- Built GraphRAG retrieval pipeline enabling natural-language search over production planning documents.

SKILLS

ML	PyTorch, Hugging Face Transformers, scikit-learn, ONNX
Programming	Python, C++, JavaScript, Git, Docker, AWS, Google Cloud
Languages	English C1, IELTS 8.0 | German C1, DSH-II | Russian native